TASK TIME DETECTIVE

LMU-STADS

WHAT SHAPES THE TOTAL DURATION

1 · Several staff work in parallel

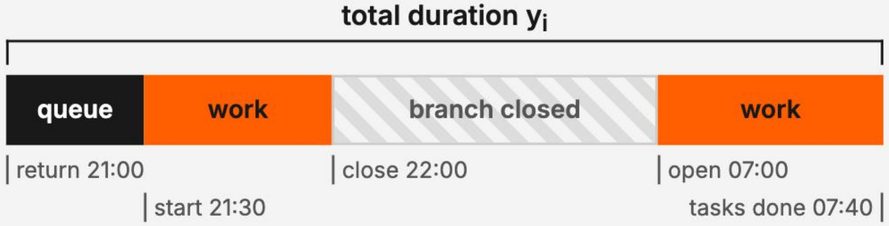
Depending on the branch, 1–3 staff work on different task chains at the same time. Tasks are done when the busiest person finishes.



Same car, same tasks: 1 person needs the sum, 3 people need only the longest chain.

2 · Queues and closed hours add time

A returned car often waits in a queue behind earlier cars before anyone starts on it. The total also runs on the clock: work pauses when the branch closes, so a car returned late carries the whole night.



Queue: grows with backlog, overnight drop-offs and daily load. Clock: opening hours per branch and weekday (Sundays closed at city centers).

⇒ $\mathrm{work}_i(\beta)$: **time until all tasks are completed**

⇒ $\mathrm{wait}_i(\gamma)$: **waiting time before work on tasks starts**

02 · METHOD

FROM TOTAL DURATIONS TO TASK TIMES

1 Structural fit

$$\min_{\beta,\,\gamma} \frac{1}{|S|} \sum_{i \in S} \left| y_i - \left(\mathrm{wait}_i(\gamma) + \mathrm{work}_i(\beta) \right) \right|$$

Powell's method optimizes β so the rebuilt totals match the observed total durations.

2 EM

E-step: each task gets its share of the unexplained time

$$\underbrace{\hat{d}_{it}}_{\text{new duration}} = \underbrace{\mu_{it}}_{\text{predicted}} + \underbrace{s_{it}}_{\text{share}} \cdot \underbrace{g_i}_{\text{unexplained}}$$

M-step: re-estimate each task formula

$$\beta_t = \arg\min_{\beta} \sum_i \Big(\underbrace{\hat{d}_{it}}_{\text{new duration}} - \underbrace{x_{it}^{\top}\beta}_{\text{formula}} \Big)^2$$

Tasks that vary more get a larger share; after 30 rounds we get the estimated coefficients β.

3 Waiting model

$$\gamma_k = \arg\min_{\gamma} \sum_{i:\, k_i = k} \left| w_i - z_i^{\top}\gamma \right|, \qquad w_i = \max\left(0,\ u_i - \mathrm{work}_i(\beta)\right)$$

A median regression per branch type predicts the waiting time before work on tasks starts.

→ Result

Coefficients β (single task durations) and γ (waiting time), utilized to estimate the overall time of the car being processed.

HEADLINE NUMBERS

PER-TASK WAPE

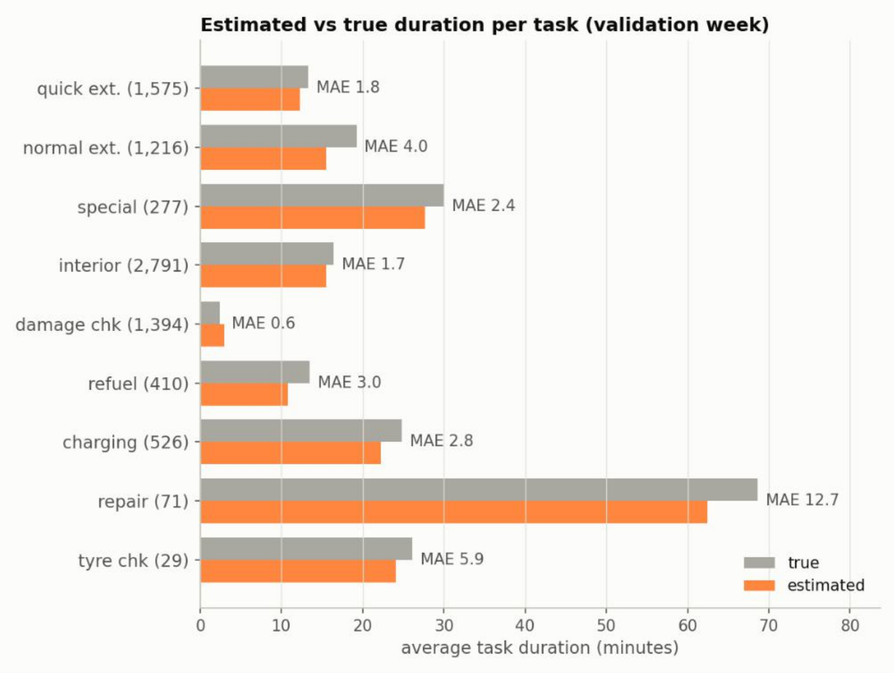
0.14

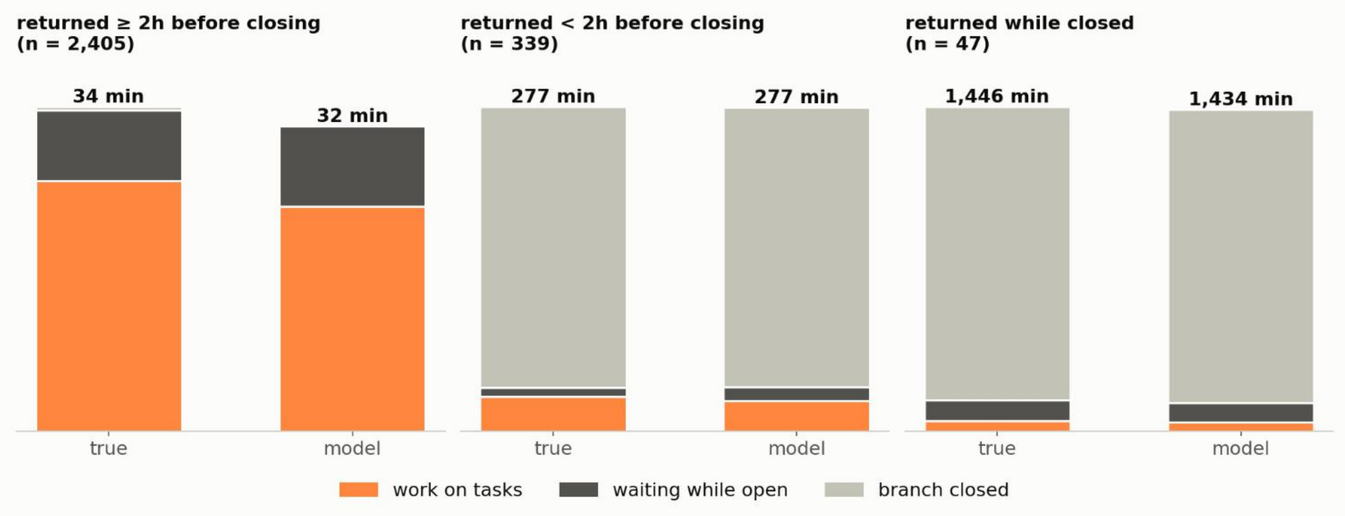
TOTAL MAE

13.6 min

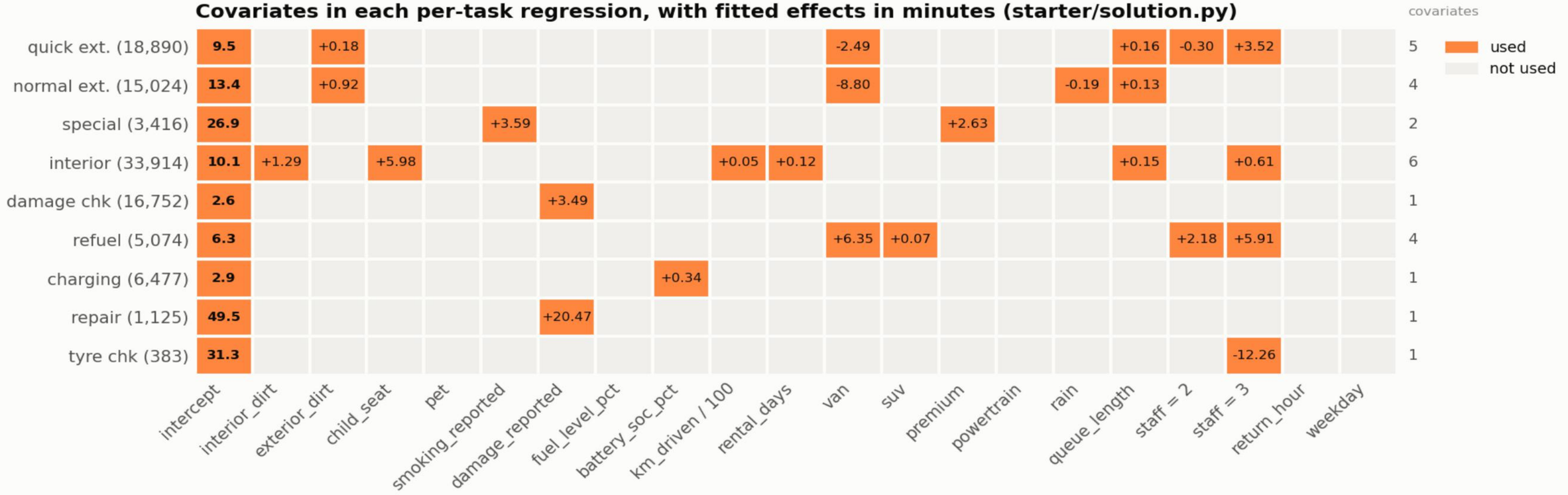
AVERAGE TASK MAE

3.87 min





FITTED EFFECTS PER TASK, IN MINUTES



Values: intercept = base minutes; others = minutes added per unit (dirt score, flag = 1, queued car, % battery missing). battery_soc_pct enters as 100 − SOC (missing → 50). staff = 2 / 3 are dummies against 1 staff (= station / airport vs centre). Count excludes the intercept. Each duration is clipped at 0.5 min.
(n) after the task = turnarounds in train.csv where the task was performed (the rows its regression is fitted on).

APPENDIX

FORMULAS AND FITTING PIPELINE

APPENDIX A · OBJECTIVE FUNCTION

REBUILT TOTALS MUST MATCH REAL TOTALS

1 · Duration of task t for car i

$$d_{it} = f_{it} \cdot \max\left(0.5,\ x_{it}^{\top}\beta_t\right)$$

Task t's duration, linear in its covariates, at least 0.5 min.

2 · Time until all tasks are completed

$$\mathrm{work}_i(\beta) = \max\left(d_{i,\mathrm{charge}},\ \min_{a \in \mathcal{A}(k_i)} \max_{p} \sum_{c:\, a(c)=p} \sum_{t \in c} d_{it}\right)$$

Best split a of task chains c over staff members $p \le k$; charging in parallel.

3 · Waiting time before work on tasks starts

$$\mathrm{wait}_i(\gamma) = \max\left(0,\ z_i^{\top}\gamma_{k_i}\right)$$

One model per staff level k: center, station, airport.

4 · Predicted total duration

$$\hat{y}_i = \mathrm{wait}_i(\gamma) + \mathrm{work}_i(\beta)$$

Counted in the branch's opening hours.

5 · Fitting objective: mean absolute error of the total

$$\min_{\beta,\,\gamma}\ \frac{1}{|S|} \sum_{i \in S} \left| y_i - \left(\mathrm{wait}_i(\gamma) + \mathrm{work}_i(\beta)\right) \right|$$

y_i: total duration

S: sample of 8,000 training cars

$\mathrm{wait}_i(\gamma)$: simplified in this fit

absolute error: robust to long waits

FROM TOTALS TO A FIRST FIT

0 Select covariates

$$j \in \text{covariates}(t) \iff |r_{tj}| \geq 0.1 \;\text{ and }\; p_{tj} < \frac{0.05}{m}$$

Consider a covariate (car/branch-related) only if it has a significant effect on task duration.

1 Starting values

$$\beta_{t,0} = \text{median}(y_i \mid f_{it} = 1) - \text{median}(y_i \mid f_{it} = 0)$$

Each task duration is initialized as the median total with the task minus without it.

2 Structural fit

$$\min_{\beta,\,\gamma} \; \frac{1}{|S|} \sum_{i \in S} \Big| y_i - \Big(\text{wait}_i(\gamma) + \text{work}_i(\beta)\Big) \Big|$$

The structural equation is optimized with Powell's method, so β gives plausible totals.

FROM A FIRST FIT TO TASK TIMES

3 EM

E-step: each task gets its share of the unexplained time

$$\hat{d}_{it} = \mu_{it} + \frac{\sigma_t^2}{\sum_{s \in \mathcal{C}_i} \sigma_s^2 + \sigma_{\text{wait}}^2}\, g_i$$

M-step: re-estimate each task formula

$$\beta_t = \arg\min_{\beta} \sum_{i:\, f_{it}=1} \left(\hat{d}_{it} - x_{it}^{\top}\beta\right)^2$$

Unexplained time is split over critical-path tasks and waiting, then each task is refit.

4 Waiting model

Median regression per staff level

$$\gamma_k = \arg\min_{\gamma} \sum_{i:\, k_i=k} \left| w_i - z_i^{\top}\gamma \right|$$

Observed waiting time

$$w_i = \max\left(0,\ u_i - \text{work}_i(\beta)\right)$$

A median regression per branch type predicts the time a car waits before work starts.

→ Predict

Predicted task duration

$$d_{it} = f_{it} \cdot \max\left(0.5,\ x_{it}^{\top}\beta_t\right)$$

Predicted total duration

$$\hat{y}_i = \text{wait}_i(\gamma) + \text{work}_i(\beta)$$

Task formulas give the work time, the waiting time is added, counted in opening hours.